



Project Guidelines

Week 8

31256 Image Processing and Pattern Recognition

42177 Image Processing and Pattern Recognition

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Assessment components

Assessment Task 1: Project Requirements and Specifications: a description of the problem, the dataset, the techniques used to tackle it (approx. 4500 words): 25%, done as a group

Assessment Task 2: Project Implementation: Report on the implementation of the project with Matlab and toolboxes (approx. 4500 words): 50%, done as a group

31256 ONLY: Assessment Task 3: Project Presentation: A presentation with PPT slides where each member of the group describes a part of the project: 25%, done as group. This is presented as both slides and a video

42177 ONLY: Assessment Task 3: Project Research Presentation: A critical review of an academic research paper related to the topic studied in the project. The student is expected to critically appraise the work in terms of the techniques applied, evaluation methods and experimental results based on knowledge obtained in the subject: 25%, done individually. This is presented as both slides and a video



Project Requirements and Specifications

From the Subject Outline:

This assignment will require each group of 5-6 students to produce a written document containing requirements and specifications for an image processing and pattern recognition project. The project topic should be granted approval by the lecturer. The specifications must cover:

- analysis of the problem, including the availability of one or more suitable, public datasets;
- selection of the techniques that the students expect to solve the problem. This will include various, plausible alternatives;
- motivation for the approach based on the knowledge acquired during lectures and independent readings;
- a performance evaluation methodology which will be used to assess the approach's performance. This will require conducting experiments on the chosen dataset(s) with the various alternative techniques and collate their performance;
- a list of references;
- the contribution of each student to the group will be described in a short table in the report.



The problem

Potential choices:

- Image classification
- Object detection
- Action recognition/detection
- Action segmentation
- Tracking
- Face detection
- Face recognition
- Others (must be approved by teaching staff)



Image classification

- Image classification: assign a class label to an image. The sets of classes can have various meanings:
 - The main *nature* of the image
 - The main *action* depicted in the image
 - The main type of *scene*
 - The main action(s) that *could be performed* in that place
- Examples of corresponding datasets:
 - ImageNet <http://www.image-net.org/>
 - Stanford-40 <http://vision.stanford.edu/Datasets/40actions.html>
 - Places <http://groups.csail.mit.edu/vision/SUN/>
 - SUN Action Dataset <http://www.di.ens.fr/willow/research/actionsfromscenes/>



Object detection

Object detection: a detection problem differs slightly from a classification problem:

- Classification: here's an object, assign it to a class
- Detection: is there an object?
- Typically, a detector is built for each type of object that we wish to detect: cars, people, horses, computers, desks, chairs (, any)
- A detector is like a two-class classifier, e.g. “classify this image patch as car/not car”
- Typically, detection and *localisation* are done simultaneously



Object detection: example





Object detection: datasets

- The most popular datasets for object detection are the PASCAL VOC challenges. They can be used for a variety of tasks including object classification, detection and segmentation, and (static) action recognition

<http://host.robots.ox.ac.uk/pascal/VOC/>

- For object classification, you can also see Caltech 101 and Caltech-256:

<https://www.kaggle.com/datasets/jessicali9530/caltech256>



Action recognition

- The word “recognition” is often used to mean: choice of features + classification
- Action recognition in video: assign an action label to a video
- Examples of datasets:
 - Hollywood 2 <http://www.di.ens.fr/~laptev/actions/hollywood2/>
 - Olympic Sports <http://vision.stanford.edu/Datasets/OlympicSports/>



Action segmentation (and classification)

This time, the video contains an uninterrupted sequence of actions: the task is to find beginning/end frames and assign a label to each

Examples of datasets:

CMU-MMAC <http://kitchen.cs.cmu.edu/>

Actions for Cooking Eggs: from Kinect <http://www.murase.m.is.nagoya-u.ac.jp/KSCGR/>

Accuracy evaluation: frame-level action labels, or accuracy of beginning/end frames and segment-level action labels



Tracking

Track single objects (people or other) in a video clip

Performance evaluation: requires dedicated measurements:

- Princeton Tracking Benchmark: a benchmark with depth and RGB video clips and with an evaluation website <http://tracking.cs.princeton.edu/>



Face detection and recognition

Face detection: just a specialised case of object detection

Face recognition: a completely different task: should be called *recognise a person's identity from an image of their face*

Like classification, but one class per person! Challenges increase with the number of people

Databases from the Face Recognition Homepage: www.face-rec.org/databases/



Techniques

You must choose plausible techniques

An example for action recognition from a video acquired with Kinect:

You could compute SIFT or STIP or other descriptors over the whole video and encode them as a Bag of Words (or a Fisher Vector or VLAD – see VLFeat); then, use the encoded vector as input into a classifier

Instead of local features, you can use the skeletal information (i.e., position of 20 joints) provided by Kinect for each frame



Evaluation

You must choose an appropriate evaluation methodology:

- For classification, accuracy, DR/FAR, precision/recall, F measure, ROC area...
- For segmentation, accuracy of the detected beginnings/ends
- For joint segmentation and classification, a combination of the above or accuracy frame-by-frame
- For tracking, MOTA/MOTP or the figures automatically computed by the PTB website
- In general: the evaluation used in the papers using the datasets

One (largish) dataset is enough, but two or three are better



References

This project requires some unspecified amount of autonomous readings

Relevant readings can be found on the public web or in repositories to which UTS is subscribed (IEEE Xplore, ScienceDirect, ACM Digital Library etc)

You must report the list of your references at the end of your Project Requirements and Specifications and cite them in the text



Project Implementation

Project Implementation Report must cover:

- Introduction to the problem
- Overview of the project and methodology
- Project Management and Team Work
- Description of the evaluation methods (datasets used, evaluation metrics, results)
- Discussion of results and challenges
- Description of the contributions of each team member
- Appendix containing a listing of the code with comments



Marking scheme – Project Requirements and Specifications

Analysis of the problem The team should clearly describe the problem that the students are studying, it's importance to society as well as the support for solving the problem. This includes the availability of one or more suitable, public datasets if the team doesn't intend to collect the data themselves.	5 pts
Selection of the techniques that the students expect to solve the problem or model the data. 1. What techniques have students identified that might be applied to solve this problem? These should be reasonably described with possible alternatives considered. 2. Motivations for the choice of the possible alternative methods should be described clearly. 3. If the project involves collecting data, then what is the proposed methodology for the data collection? 4. If the project involves the use of publicly available datasets, then the report should identify suitable, public datasets and describe why these datasets are appropriate for the project.	5 pts
Performance Evaluation Methodology The team should be able to describe clearly how they will evaluate and compare the different methods for solving the problem and what a successful outcome should look like considering the common applications of the problem they are investigating.	5 pts
Reference List A list of appropriate references to show that the team have studied the problem in detail with the references correctly cited and discussed in the report.	5 pts
Professionally Written Report The report should be formatted in a professional manner with an Introduction, Conclusion, list of references (appropriately referred to in the main text), figure captions and correct citation of external sources. In addition, there should be an absence of spelling or grammar errors. The report should contain a table detailing the contributions of each team member to the activity.	5 pts
Total Points: 25	

Marking scheme – Project Implementation



Introduction to the Problem This should describe the project, motivations and techniques used in a re-cap of the Project Requirements and Specifications Report. It does not need to be as detailed as the Project Requirements and Specifications Report, but should provide sufficient information for the reader to understand the nature of the project without needing to refer directly to the Project Requirements and Specifications Report. It should not be a direct copy of the Project Requirements and Specifications.	5 pts
Overview of the project and methodology The report should describe methods and techniques that the team used to achieve the results shown in the later parts of the report. Results should not be presented here. This section should give an overview of the project and the techniques used. In this section, the report should answer the following questions: 1. What did the team do to achieve the results shown later in the report? What techniques did they use? 2. If the team collected the data themselves, then what was the experimental setup to collect the data? 3. If the team used publicly available datasets, then what was the nature of the datasets used?	10 pts
Project Management and Team Work The team should describe how the team worked together on the project and how they divided tasks between the team members. This section should include a table indicating the contribution of each team member to the overall project.	5 pts
Results The report should describe the results that the team obtained. The data should be visualised in a compelling manner that demonstrates clearly the insights that the team discovered from the data or the effectiveness of models produced by the team.	10 pts
Discussion and Challenges The report should insightfully discuss what the students learnt from this project, including: 1. Where the students able to achieve their initial goals when starting the project? Why or why not? 2. What challenges did the students face during the project? How did they change or adjust their plan to account for the challenges? 3. What are their thoughts on future work in this area? What would they have done if given more time? 4. What did they learn about the problem from this exercise?	10 pts
Professionally Written Report The report should be formatted in a professional manner with an Introduction, Conclusion, list of references (appropriately referred to in the main text), figure captions and correct citation of external sources. In addition, there should be an absence of spelling or grammar errors.	5 pts
Appendix with code The report should have an appendix with the code created. The code should be clearly and professionally commented.	5 pts
Total Points: 50	



Marking scheme – 31256: Project Presentation

Presentation Quality Slides are well designed and engaging. Figures and charts clearly demonstrate the concepts.	7 pts
Presentation is Consistent with Project Report Presentation clearly covers results, challenges and key discussion points of the project report.	5 pts
Engaging Presentation by Team Members Team members take effort to clearly identify themselves and speak and present in an engaging manner.	6 pts
Team Members show Good Knowledge of the Work Each team member shows a good understanding of the part of the presentation that they are responsible for. Each team member has a component of the talk to present and clearly gives their name at the beginning of their part of the presentation.	7 pts
Total Points: 25	



Marking scheme – 42177: Project Research Presentation

Presentation Quality Slides are well designed and engaging. Figures and charts clearly demonstrate the concepts. The speaker conveys the concepts in an engaging fashion	7 pts
Description of the work Presentation clearly describes the science and technical detail of the reviewed work in a manner that is clear to the audience. The presentation should describe clearly the key contribution of the work and what makes the work a useful contribution to the field. If the presenter believes that the work falls short, then reasons should be clear and convincing.	6 pts
Critique of the Experimental Design Presentation describes clearly the experimental design and why this design is or is not appropriate to the problem. The presentation should detail flaws or particular advantages of the experimental design.	6 pts
Overall Judgement of the Work The presentation should discuss the critical aspects of the work including particular challenges or flaws in the work and future work. The presenter should give an opinion on the overall quality of the reviewed work, how it could be improved and what additional work is needed.	6 pts
Total Points: 25	



Submission

All assessment items must be submitted via Canvas by one of the group members by the due date

A link is available for each assessment item; if you have problems, just please contact me by email

for Assessment Item 1, you must submit a document (MS Word or PDF);

for Assessment Item 2, you must submit a report on your code including an appendix with a well-commented listing of the code.

for Assessment Item 3, you must submit your slides (PPT or PDF) and a video